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Stall AI-First D&M Platform

Purpose/Objective:	This document describes the complete intended Project 1 product. It is not an MVP definition and does not intentionally reduce the product to a small validation release. The scope covers the end-to-end AI-first workflow and outlines the strategic vision & functional direction in a narrative format for the product being built for Vasant Creative Ltd.
Recipient:	Mrs. Vidhi Vikamsey
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POC:	Mrs. Vidhi Vikamsey
Overview:	This document provides a clear, narrative-level explanation of the vision and purpose behind the Stall AI-First D&M Platform. It focuses on the following key areas as requested: • ***What the product is designed to achieve*** • ***The core problem the product is solving*** *the practical need addressed through the platform.* • ***The expected end-to-end experience for key users*** *including designers, project managers, procurement teams, warehouse teams, and clients.* • ***What a "successful" product looks like*** *from both a business outcome and user-experience standpoint.* • ***What the product is fundamentally meant to do :*** *written as a narrative rather than a technical specification.*

This document has been prepared based on the scope received and the follow-up clarifications provided by the Vasant Creative Ltd. team.

This document has been purposefully created at the product and experience level. Its objective is to establish a clear understanding of the proposed project, its intended capabilities, and the overall outcome it is expected to deliver. It does not intend to define technical terminology, system architecture, architectural blocks, implementation methodology, or underlying technology at this stage.

The purpose of this stage is to mutually understand, discuss, refine and approve the proposed feature set and overall product scope. Once the features and scope are mutually agreed upon, the subsequent technical scoping process will begin, where the approved requirements will be translated into the relevant technical architecture, processes, development stages, integrations and implementation requirements, followed by detailed costing.

We hope this document provides the required clarity and serves as a strong foundation for mutually aligning on the proposed product.

Thank you.

Best regards,
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1. Product Vision

Project 1 is an AI-native platform for designing, planning, costing, visualising and managing exhibition stalls. The product is designed around a simple shift in workflow: instead of beginning with a blank design application and manually constructing a booth from scratch, users begin with a brief, brand, space, inventory, budget and functional requirements.

The platform interprets those requirements, builds a structured booth representation, generates design concepts, allows the team to iterate conversationally, and turns the approved concept into usable spatial, visual, commercial and production-oriented outputs.

The vision is simple:

To create the most adaptive, realistic and personalised exhibition stall design environment where every project can move from brief to a complete, buildable design.

2. Conceptual Flow

2.1. Project Intake & Requirement Structuring

SR No. / IE: (Refer to Section 3.)
1 — A1 · 2 — B1 · 3 — B2 · 4 — C1

The project begins by establishing a complete structured project brief. The user provides the exhibition and event details, client and brand information, project reference, booth footprint, dimensions, height limitations, access points, structural obstructions, fire exits, mandatory clearances, venue context and functional requirements.

Users can upload reference images, previous designs, sketches, PDFs, brand guidelines and other project-specific material. Client brand information including logos, colours, typography, visual identity and brand references is captured as part of the project context.

The platform converts these inputs into a structured requirement checklist and applies the defined physical and functional constraints before design generation begins.

2.2. Enterprise Resource & Commercial Data Integration

SR No. / IE:
12 — H1 · 13 — H2 · 14 — H3 ·
15 — H4 · 16 — I1 · 17 — H5

The platform connects the project with Vasant Creative's personalised RAG storage which will hold warehouse inventory, materials costing and vendor ecosystem.

Existing furniture, structures, display units and reusable assets can be connected directly to the project. Inventory can also be manually added with images, dimensions, categories, availability and other attributes.

The platform identifies whether assets are available, allocated or unavailable and allows selected inventory to be associated with a specific project or design. When an item is unavailable, the platform will suggest suitable alternatives from available company inventory.

2.3. Persistent Project Context & Retrieval-Augmented AI

SR No. / IE:
19 — K1 · 20 — K2 · 21 — L1 ·
22 — D3 · 30 — L2 · 52 — P1 · 59 — A4

The platform operates through a persistent model workspace rather than isolated prompts. The model maintains awareness of the project brief, brand, dimensions, references, inventory, materials, budget and decisions throughout the project lifecycle.

Relevant information can be retrieved from previous Vasant Creative projects and the central company knowledge repository, including approved materials, inventory, vendors, pricing, historical designs and other reusable information.

The model retains approved decisions, rejected concepts, project preferences and design instructions so users do not have to repeatedly explain the same requirements.

Global search allows users to discover information across projects, designs, inventory, materials, vendors and company knowledge. Authorised users can also control model rules, approved data sources, design constraints and company-specific model behaviour.

2.4. Generative Concept Synthesis & Multi-Direction Exploration

SR No. / IE:
23 — D4 · 24 — D5 · 25 — D6 ·
33 — M1 · 34 — N1 · 35 — N2 · 11 — A3

Once the project context is established, the model generates initial stall design concepts using the brief, brand identity, booth dimensions, functional requirements, budget and available company resources. Multiple distinct design directions can be generated from the same project brief, allowing the team to explore different layouts, visual languages and spatial approaches before selecting a preferred direction.

Uploaded references, previous designs and approved project material can be used as controlled inputs during generation. Budget, available inventory and approved materials can also influence the generated concepts.

Different concepts or versions can then be compared based on visual direction, functionality, budget and project requirements before the team selects the direction to develop further.

2.5. Modular Spatial Generation & Design Composition

SR No. / IE:
9 — A2 · 10 — G1 · 26 — D7 ·
31 — D11 · 43 — B3 · 44 — G3

Large exhibition stalls are developed as independent blocks or segments instead of requiring the complete stall to be generated as one image.

Individual areas such as entrances, lounges, meeting spaces, product displays or other booth zones can be generated and iterated independently while the model maintains awareness of the overall design language, approved elements, dimensions, materials and project context.

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Finalised blocks are automatically stitched into a continuously developing master representation of the complete stall.

Multiple team members can work on different blocks simultaneously, while a live stitched preview shows how the individual sections combine into the overall stall. Version history maintains previous generated, modified and approved states so earlier design versions can be reviewed or restored.

2.6. Visual Annotation & Targeted Regeneration

SR No. / IE:
27 — D8 · 28 — D9
29 — D10 · 31 — D11 · 32 — C2

Designers can interact directly with generated visuals by identifying a specific area and communicating exactly what needs to change. A selected area can be annotated using visual or textual instructions, allowing the model to regenerate only the required portion instead of recreating the entire design. Approved elements should remain protected during regeneration, while negative instructions allow users to explicitly define elements that must not be introduced, removed or modified.

The model maintains consistency of branding, furniture, materials, dimensions, design language and approved elements across the revised design.

Generated designs can subsequently be validated against the project's dimensions, requirements, physical constraints, budget, inventory and other defined project rules.

2.7. Spatial Semantic Modeling & Technical Design Generation

SR No. / IE:
5 — D1 · 6 — D2 · 7 — E1
· 8 — F1 · 55 — Q1

Once a visual design is approved, it moves beyond being a rendered image and becomes a structured spatial representation. The system interprets booth geometry, zones, objects, dimensions and relationships between elements. This structured understanding can then be used to generate top views, plans, elevations and structured 3D representations.

Major booth elements such as walls, counters, furniture and zones can have measurable dimensions within the spatial model.

Users can explore the resulting 3D booth from different viewpoints and perspectives. Where supported by the final technical scope, this structured representation can also form the basis for production-oriented technical outputs.

2.8. Parametric Cost Intelligence & Budget Optimisation

SR No. / IE:
36 — D12 · 37 — O1 · 38 — O2
39 — O3 · 40 — O4 · 41 — O5 · 54 — O6

Cost remains connected to the design process rather than being calculated only after the design is complete. The costing engine calculates project cost using the design, materials, inventory, vendors,

quantities and other commercial inputs. Estimated cost can update as designs are generated, modified or finalised.

Different design scenarios can be compared based on total cost, material usage, inventory usage and other commercial variables. Budget alerts identify when a design or modification exceeds defined thresholds.

The model can also modify and optimise a design towards a target budget while attempting to preserve the intended design direction, functionality and visual quality.

Once the design is approved, quantities, materials and components can be converted into BOQ/BOM information and the corresponding commercial package.

2.9. Collaborative Review, Client Interaction & Approval Governance

SR No. / IE:
42 — G2 · 45 — G4 · 46 — G5 · 47 — G6 · 49 — G8 · 50 — G9 · 51 — G10 · 60 — G14

The platform provides a shared project workspace where multiple designers and team members can work around the same project model context and design.

Internal stakeholders can review designs while clients or authorised external stakeholders can access a simplified presentation environment without exposure to internal project information.

Clients and stakeholders can provide comments on complete designs or specific areas and request revisions. These interactions feed back into the design workflow.

An approval workflow manages review, revision requests and approved versions. Role and access management controls what management, designers, project managers, clients and other users can access.

Project activity and audit trails retain important design changes, approvals, revisions and other significant actions, while notifications keep relevant users informed about approvals, assignments, completed generations, deadlines and revisions.

2.10. Approved Design Output & Production Package Orchestration

SR No. / IE:
53 — B4 · 54 — O6 · 55 — Q1 · 56 — G11 · 57 — G12

After approval, the platform converts the final project state into usable design and commercial deliverables.

The design package can contain approved visualisations, plans, elevations, 3D views and other agreed design outputs.

The commercial package can contain costing, BOQ/BOM and other commercial information derived from the approved structured design.

Where defined within the final technical scope, the structured spatial design can also support production-oriented outputs.

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Approved information can be exported in agreed standard formats, while structured company information such as inventory, materials, vendors and pricing can be imported or exported for operational use.

2.11. Institutional Knowledge Capture & Project Reuse

SR No. / IE:
19 — K1 · 20 — K2 · 21 — L1 · 30 — L2 · 61 — P2*

Completed projects become reusable company knowledge rather than remaining isolated project files.

The platform can retain project designs, decisions, costing, materials, inventory usage, references and other relevant information within Vasant Creative's knowledge environment.

Completed projects can be archived while retaining the information required for future retrieval and reuse. Future projects can therefore benefit from previous design approaches, historical pricing, materials, inventory usage and company knowledge.

This allows the model to continuously build upon Vasant Creative's accumulated design and commercial experience rather than treating every new project as an entirely new problem.

Note: Project 1 is a full AI-first generative design platform.
It is intentionally broad and production-oriented, but it is not intended to reproduce every manual capability of professional CAD/3D design software.

Note: Following Scope is NOT a MVP.
The scope described here is the complete intended Project 1 product. Individual implementation phases may exist during delivery, but the product definition itself should not be reduced to a small demonstration or validation release.

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3. Comprehensive Features Stack

The following feature stack represents the proposed and complete intended product flow. It is organised around how Vasant Creative would actually use the website rather than around isolated technical modules.

Please annotate the final column as Required / Not Required based on the requirements of the final product.

#	IE	Feature	Feature Description	Layer	Annotate
1	A1	Project – Requirement Checklist	Convert the project brief into a structured checklist of mandatory requirements, spaces, branding elements, furniture, utilities and functional needs.	Model / Platform	Yes / No
2	B1	Project – Reference Uploads	Allow users to upload reference images, previous designs, sketches, PDFs, brand guidelines and other project-specific material.	Platform / AI	Yes / No
3	B2	Project – Brand Information	Capture client brand information including logos, colours, typography, brand guidelines, references and other visual identity information.	Platform / AI	Yes / No
4	C1	Model – Design Constraints	Apply defined constraints such as booth dimensions, height limits, access points, columns, fire exits, mandatory spaces and other physical or functional limitations.	Model / Spatial	Yes / No
5	D1	Model – Spatial Design Understanding	Convert generated designs into a structured understanding of booth geometry, zones, objects, dimensions and spatial relationships.	Model Engine	Yes / No
6	D2	Model – 2D / 3D Generation	Generate top views, plans, elevations and structured 3D representations from the approved spatial design.	Model Engine	Yes / No
7	E1	Spatial – Measurement System	Provide measurable dimensions for walls, counters, furniture, zones and other major booth elements within the structured design.	Spatial Engine	Yes / No
8	F1	Spatial – View Control / Walkthrough	Allow users to explore the generated 3D booth from different viewpoints and perspectives.	3D / Platform	Yes / No
9	A2	Design – Block Stitching	Automatically stitch finalised design blocks into a growing complete stall representation while maintaining continuity with previously approved blocks.	Model / Platform	Yes / No
10	G1	Design – Version History	Maintain versions of generated, modified and approved designs so previous design states can be reviewed or restored.	Platform	Yes / No
11	A3	Design – Comparison	Compare multiple design concepts or versions based on visual direction, functionality, budget and project requirements.	Model / Platform	Yes / No
12	H1	Inventory – Warehouse Integration	Connect Vasant Creative's warehouse and inventory data so reusable furniture, structures, assets and materials can be considered during design generation.	Company Data	Yes / No
13	H2	Inventory – Manual Upload	Allow inventory items to be manually added with images, dimensions, categories, availability and other relevant attributes.	Company Data	Yes / No
14	H3	Inventory – Availability Check	Identify whether selected furniture, assets and materials are currently available, allocated or unavailable.	Company Data	Yes / No
15	H4	Inventory – Design Reservation	Associate selected inventory items with a project so assets can be tracked against a particular design or project.	Company Data	Yes / No
16	I1	Inventory – Alternative Suggestions	If a selected inventory item is unavailable, suggest suitable alternatives from available company inventory.	Model / Inventory	Yes / No
17	H5	Materials & Vendor Database	Maintain material, vendor and supplier information including specifications, categories, availability and actual commercial pricing.	Company Data	Yes / No

18	J1	Brand – Web Scraping	Automatically collect relevant publicly available client brand information, references and assets from the client's website or approved online sources.	Model / Data	Yes / No
19	K1	Knowledge – Previous Projects	Allow previous Vasant Creative projects, designs, materials, pricing and related information to be reused for future work.	Company Data / RAG	Yes / No
20	K2	Knowledge – Central Repository	Maintain a structured company knowledge base containing approved materials, inventory, vendors, previous projects, pricing, design references and other reusable information.	Company Data / RAG	Yes / No
21	L1	Knowledge – RAG Context	Pull relevant project, company, inventory, material, vendor and historical information into the Model context when generating or modifying designs.	Model / RAG	Yes / No
22	D3	Model – Project Conversation	Provide a persistent project-specific Model conversation based upon RAG survey context insuring a 100% accuracy and less than 1% of hallucination of the model.	Model Engine	Yes / No
23	D4	Model – Design Concept Generation	Generate initial stall design concepts based on the project brief, brand, dimensions, functional requirements, budget and available company assets.	Model Engine	Yes / No
24	D5	Model – Multi-Concept Generation	Generate multiple distinct design directions from the same brief so the team can compare and select a preferred concept.	Model Engine	Yes / No
25	D6	Model – Reference-Aware Generation	Use uploaded reference images, previous designs and approved project references as controlled inputs during design generation.	Model Engine	Yes / No
26	D7	Model – Block-by-Block Generation	Generate a large stall portion by portion while maintaining consistency with the overall design and project context.	Model Engine	Yes / No
27	D8	Model – Design Iteration	Allow designers to modify and regenerate selected design elements while preserving approved elements and maintaining overall design consistency.	Model Engine	Yes / No
28	D9	Model – Annotation	Allow designers to mark a specific area of a generated design and instruct the Model on exactly what should be changed.	Model Engine	Yes / No
29	D10	Model – Negative Instructions	Allow users to explicitly define elements that should not appear or should not be changed during regeneration.	Model Engine	Yes / No
30	L2	Model – Design Memory	Retain approved decisions, rejected concepts, project preferences and design instructions throughout the project lifecycle.	Model / RAG	Yes / No
31	D11	Model – Design Consistency	Maintain consistency of materials, furniture, branding, dimensions, design language and approved elements across multiple generated views and blocks.	Model Engine	Yes / No
32	C2	Model – Design Validation	Validate generated designs against defined dimensions, requirements, constraints, budget, inventory and other project rules.	Model / Spatial	Yes / No
33	M1	Model – Budget-Aware Generation	Use the defined project budget and commercial information as constraints during design generation and iteration.	Model / Cost Engine	Yes / No
34	N1	Model – Inventory-Aware Generation	Prefer or incorporate available company inventory and reusable assets where appropriate during design generation.	Model / Company Data	Yes / No
35	N2	Model – Material-Aware Generation	Generate designs using defined materials and material properties available within the approved material database.	Model / Company Data	Yes / No
36	D12	Model – Cost Optimisation	Help modify the design to achieve a target budget while maintaining the intended design direction, functionality and visual quality.	Model Engine	Yes / No
37	O1	Costing – Engine	Calculate project cost using the design, materials, inventory, vendors, quantities and other commercial inputs.	Cost Engine	Yes / No
38	O2	Costing – Live Cost	Display live estimated project cost during design generation, modification and block finalisation.	Cost Engine	Yes / No
39	O3	Costing – Scenario Comparison	Compare different design versions based on total cost, material usage, inventory usage and other commercial variables.	Cost Engine	Yes / No

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40	O4	Costing – Budget Alerts	Alert users when a design or design change exceeds the defined project budget or crosses defined cost thresholds.	Cost Engine	Yes / No
41	O5	BOQ / BOM – Generation	Generate quantities, materials, components and other bill-of-quantity / bill-of-material information from the approved structured design.	Cost Engine	Yes / No
42	G2	Collaboration – Workspace	Allow multiple designers or team members to work within the same project workspace and collaborate around the project Model conversation and design.	Platform	Yes / No
43	B3	Collaboration – Parallel Block Generation	Allow different team members to work on different stall blocks while maintaining shared project context and ultimately combining those blocks into the complete design.	Platform / AI	Yes / No
44	G3	Collaboration – Live Stitched Preview	Display the continuously updated stitched stall design while individual team members generate and finalise different sections.	Platform	Yes / No
45	G4	Client – Presentation Mode	Provide a simplified client-facing view for presenting approved concepts without exposing internal project information.	Platform	Yes / No
46	G5	Client – Feedback & Comments	Allow authorised clients or stakeholders to comment on designs or specific areas and request revisions.	Platform	Yes / No
47	G6	Approval – Workflow	Manage design review, revision requests, approvals and approved versions across relevant stakeholders.	Platform	Yes / No
48	G7	Project – Dashboard	Provide management with an overview of active projects, project status, deadlines, assigned users, approvals and team activity.	Platform	Yes / No
49	G8	Project – Role & Access Management	Provide different access levels for management, designers, project managers, clients and other users according to their role.	Platform	Yes / No
50	G9	Project – Activity & Audit Trail	Maintain a record of important project actions, design changes, approvals, revisions and other significant activities.	Platform	Yes / No
51	G10	Notifications – Project Updates	Notify relevant users about approvals, revision requests, completed generations, assignments, deadlines and other important events.	Platform	Yes / No
52	P1	Search – Global Search	Search across projects, designs, inventory, materials, vendors, previous projects and other company knowledge.	Platform / RAG	Yes / No
53	B4	Output – Design Package	Produce agreed design outputs including visualisations, plans, elevations, 3D views and other approved deliverables.	Platform / AI	Yes / No
54	O6	Output – Commercial Package	Produce costing, BOQ/BOM and other commercial information derived from the approved design.	Cost Engine	Yes / No
55	Q1	Production – Technical Outputs	Generate agreed production-oriented outputs from the structured design, subject to the capabilities defined within the final technical scope.	Spatial / Platform	Yes / No
56	G11	Export – Standard Formats	Export approved designs, images, plans, elevations, costing and BOQ/BOM information in agreed formats.	Platform	Yes / No
57	G12	Data – Import / Export	Allow structured company data such as inventory, materials, vendors and pricing to be imported and exported in agreed formats.	Platform	Yes / No
58	G13	Admin – Company Configuration	Allow authorised administrators to configure company rules, inventory categories, materials, vendors, pricing, roles and other system-level settings.	Platform	Yes / No
59	A4	Admin – Model Controls	Allow authorised users to define Model generation rules, approved data sources, design constraints and company-specific Model behaviour.	Model / Platform	Yes / No
60	G14	Security – Role-Based Access	Control access to projects, company data, financial information, client information and administrative functions based on user roles.	Platform	Yes / No
61	R1	Data – Project Archiving	Archive completed projects while retaining their relevant designs, costing, decisions and knowledge for future reuse.	Platform / RAG	Yes / No
62	G15	Analytics – Project Performance	Provide management insights into project volume, design iterations, approval time, budgets, costs and other operational metrics.	Platform	Yes / No

63	G16	Management – Project Overview	Provide management-level visibility across projects, including project progress, estimated cost, approvals, workloads and other relevant operational information.	Platform	Yes / No

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4. Delivery Timeline

The complete Project 1 scope, including all features marked as Required, is targeted for delivery within a maximum period of 25 days, inclusive of development, integration, testing, bug fixing and final delivery.

The delivery timeline applies to the **mutually approved scope**. Any additions or material changes to the agreed feature set during development may require a revised delivery timeline and will be mutually discussed before implementation.

Positioning Statement

This document is intended to establish and mutually confirm the **Feature Stack** for Project 1. Vasant Creative Ltd. is requested to review the feature stack and annotate each feature based on the requirements of the final product.

Once the feature stack is mutually agreed upon, the project will proceed to the costing phase, where the development effort and associated costs will be determined based on the selected features.

Following the costing phase, a detailed Technical SOP will be prepared. This will translate the approved feature scope into the technical and implementation-level requirements, including the end-to-end process for each feature, individual orchestrators, RAG engines, hosting requirements, APIs, integrations, third-party services, access and permissions, product rights, and other relevant technical and product considerations.

Therefore, this document intentionally remains at the **product, feature and user-experience level** and does not define the underlying technical architecture or implementation methodology.

Please revert with the selected features by annotating the final column as Required / Not Required.
